TEACHER GUIDE

Cambridge IGCSE Business (0450 / 0264)

Stock

4 Operations management
4.1 Production of goods and services: inventory and lean production



A revision lesson. Pupils run a car dealership in firms of four and revise stock on their own deals. Seat each firm together. Each firm answers the topic tasks together.

The lesson

45-minute setting

Setup	Round 1	Round 2	Round 3	Final questions

EVERY ROUND

Firm decisions
3 min

→

Deals
4½–5 min

→

Topic tasks
4 min

→

B2B deal
2½ min
from Round 2

LESSON SETTINGS

30 min	45 min	60 min	75 min

What pupils do

Every pupil buys and sells cars. In Firm decisions, each role has its own job.



CEO

Staff · Sabotage from Round 2



Finance

Capital & Loans · Investments



Sales & Marketing

Marketing



Operations

Seasonal stock

Topic tasks

Each firm gets its own figures. The method is the same.

1 Stock holding cost

Round 1

Cars wait in your yard.

Cost of holding the cars?

You hold for them	**8 cars**
Holding cost per car a week	**£100**
Weeks you hold them	**4 weeks**

ANSWER

£3,200

8 × £100 × 4 = £3,200

Why: Stock that waits costs £800 every week.

2 Too much stock

Round 1

Your yard holds far more cars than you sell in a month.

Which two problems are most likely?

A Running out of cars to sell
B Higher storage costs
C Lower insurance costs
D Cash tied up in stock
E Lost sales from empty space

ANSWER

✓ **Higher storage costs**
✓ **Cash tied up in stock**

Why: Extra stock holds cash and costs money to keep.

3 Rank supplier cost

Round 2

Three suppliers quote for 10 cars, delivered.

Order by total cost, cheapest first?

A £8,000 a car, £2,500 delivery
B £8,100 a car, £1,200 delivery
C £8,300 a car, £300 delivery

ANSWER

1. **£8,100 a car, £1,200 delivery**
2. **£8,000 a car, £2,500 delivery**
3. **£8,300 a car, £300 delivery**

Why: £82,200, £82,500 and £83,300: the lowest price per car is not the cheapest order.

Topic tasks

4 Saving from JIT

Round 2

Just in time, or in case?

Saving a month from JIT?

Spare cars held, just in case	**7 cars**
Holding cost per car a week	**£70**
Extra JIT deliveries, a week	**£200**
Weeks in the month	**4 weeks**

ANSWER

£1,160

4 × (7 × £70 − £200) = £1,160

Why: JIT saves the holding cost, but the extra deliveries cost £800 a month.

5 JIT or hold stock

Round 3

Your supplier is two hours away and always on time. Hold 40 cars, or hold 10 and reorder often?

Which should you choose, and why?

A Hold 40 cars

B Hold 10 and reorder often

ANSWER

✔ **Hold 10 and reorder often**

Why: A reliable, nearby supplier lets you hold less and free up cash.

6 JIT vs discount

Round 3

JIT loses a bulk discount.

Gain a month from JIT? Minus allowed.

Cars bought a month	**13 cars**
Price per car	**£8,000**
Bulk discount lost with JIT	**3%**
Holding cost saved a month	**£2,000**

ANSWER

−£1,120

13 × £8,000 × 3% = £3,120;
£2,000 − £3,120 = −£1,120

Why: Smaller, frequent orders save holding costs but can lose bulk-buying discounts.

Final questions

Each pupil answers 6 final questions on stock. They come from these 14 questions.

1 What is one risk of relying on a single overseas parts supplier?
✔ **A problem abroad could stop its supply**

2 Which evidence set is most useful when selecting a supplier for regularly needed vehicle parts?
✔ **Price, quality, reliability and delivery performance**

3 What is an opportunity cost of holding too much vehicle stock?
✔ **The cash cannot be used for anything else**

4 Supplier A is slightly cheaper, but frequently delivers parts late. Supplier B is reliable. Which factor supports choosing B for urgent repairs?
✔ **Dependability of delivery**

5 A workshop keeps almost no spare parts and orders only when needed. What is the main risk of this just-in-time approach?
✔ **A late delivery can stop work**

6 Delivery fuel costs rise sharply. Which operational response is most likely to reduce the impact?
✔ **Plan shorter routes with fuller vans**

7 A key supplier enters insolvency. Which information is most important first?
✔ **Open orders, stock cover and backup suppliers**

8 High-value parts repeatedly disappear from stock records. Which control system is most useful?
✔ **Restrict access, log movements and reconcile counts with records**

9 A supply chain is every business that helps get a product to the customer. A tyre maker's factory floods. Using this, what happens to a workshop?
✔ **Its tyre deliveries may stop, though it did nothing wrong**

10 A reliable supplier delivers the right parts, in full, when promised. The cheapest supplier delivers late one time in three. What does staying with it really cost?
✔ **Delayed jobs and waiting customers can cost more than the saving**

11 What does just-in-time (JIT) stock control mean for a dealer's parts department?
✔ **Parts arrive just as they are needed**

Final questions

12 A dealer holds far more spare parts than it needs. What is one problem with this?

✔ **Cash is tied up in parts on shelves**

13 A workshop keeps very few parts in stock, and a delivery of brake pads arrives late. What is the most likely result?

✔ **Repairs are delayed**

14 Which is a disadvantage of just-in-time stock control for a dealer’s workshop?

✔ **One late delivery can stop work**